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From Planning to Execution: The Role of Drilling Fluid Engineering in Successful Combined Well Completion Operations

Alaa Zahran, Hany Gamal, Mostafa Salim, and Mohammed Atiah, Weatherford Saudi Arabia

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Abstract

The integration of off-bottom liners and lower completions in a single-run operation represents a paradigm shift in well completion strategies. Overcoming technical challenges such as differential sticking, fluid invasion, and inefficient hole cleaning in high-permeability reservoirs requires advanced engineering solutions. This study illustrates the pivotal role of drilling fluid engineering in enhancing wellbore stability, operational efficiency, and formation integrity during a single-run deployment, showcasing a robust framework for mitigating challenges and achieving significant cost and time savings.

This research adopts a systematic approach to design and manage particle size distribution (PSD) for sealing permeable formations. The methodology integrates graphical PSD optimization, enabling precise alignment of bridging material properties with formation characteristics derived from pore size and permeability data. Engineering simulations were utilized to calculate ideal particle size distributions, providing customized solutions mud system. Field deployment protocols incorporated PSD monitoring and maintenance to ensure continuous performance during dynamic drilling operations. The workflow also included recommendations for adjusting PSD to accommodate combined well completion operations, thereby maximizing sealing efficiency and minimizing fluid invasion. The drilling fluid engineering team studied high-technical mud characteristics design to ensure optimal hole-cleaning efficiency, particularly in mitigating differential sticking and fluid loss in high-permeability zones, where permeability reached up to 5000 mD.

The applied engineering workflow delivered outstanding results, setting a new standard for high-permeability drilling operations. The proposed approach was implemented in the reservoir section using a limited selection of bridging materials that were exclusively acid-soluble, avoiding any materials that could cause reservoir damage. This design maintained an overbalance pressure of approximately ± 2000 psi and handled permeability levels up to 5000 mD.

Additionally, the integration of advanced simulation models allowed for predictive optimization, ensuring alignment between field execution and pre-designed performance metrics. The combined operation was successfully executed within 24 hours, achieving a 50% reduction in operational time compared to the conventional two-run approach, which typically required 48 hours. The operation concluded with all planned activities completed without deviations, highlighting the impact of pre-job planning and operational

precision. The advanced engineering single-run approach not only delivered significant time savings but also reduced overall project costs, showcasing engineering excellence in optimizing operational workflows.

This study highlights a transformative approach in drilling fluid engineering that ensures operational success in single-run deployments. This advanced drilling fluid engineering approach to combining off-bottom liner and lower completion operations emphasizes the importance of collaborative engineering, advanced modeling, and precise execution. The single-run methodology demonstrates significant potential for reducing operational time and costs while maintaining uncompromised safety and reliability.

Key Words: Drilling Fluid Engineering, Particle Size Distribution (PSD), Differential Sticking, Off-Bottom Liner, Lower Completion, Combined Operations, Operational Efficiency, Cost Optimization

Introduction

Efficient and cost-effective well construction is a primary objective in the petroleum industry. As reservoirs become more complex and economically marginal, optimizing every phase of the well delivery process is critical. The integration of multiple completion components into a single operational run represents a significant advancement aimed at reducing rig time and associated costs. However, executing such combined operations in challenging environments, particularly high-permeability formations, introduces distinct technical hurdles (Ong et al., 2024).

Differential sticking, where the drill string becomes embedded in a permeable formation due to differential pressure, fluid invasion into the reservoir rock leading to potential formation damage, and ensuring adequate hole cleaning to prevent packing off are magnified in these scenarios (Gamal et al., 2019). Traditional drilling fluid designs and operational practices, while effective in conventional drilling, often require substantial refinement to successfully navigate the complexities of single-run combined completion deployments.

The Role of Drilling Fluid Engineering in Drilling and Well Completion Operations

Drilling fluid, often referred to as "mud," serves as the lifeblood of any drilling operation. Its multifaceted functions are essential for successful and safe wellbore construction. These functions extend beyond simply carrying cuttings to the surface; they encompass maintaining wellbore stability, controlling formation pressures, cooling and lubricating the drill bit and string, facilitating formation evaluation, and supporting the drill string and casing (Mondal and Chatterjee, 2019). The effectiveness of drilling fluids directly impacts drilling efficiency, wellbore quality, and the potential for formation damage, all of which are critical considerations in both drilling and subsequent completion operations (Goud and Joseph, 2006; Sheer et al., 2020).

Drilling fluids technically provide operational functions during the drilling program, as hole cleaning due to the fluid's ability to transport drilled cuttings from the bottom of the wellbore to the surface is paramount. This function is governed by the fluid's rheological properties (viscosity, yield point), annular velocity, and hole geometry. In deviated or horizontal wells, maintaining adequate carrying capacity is particularly challenging due to the gravity settling of cuttings. Carrying cuttings as mentioned in hole cleaning, the fluid suspends and transports cuttings. The fluid's gel strength helps keep cuttings suspended when circulation is stopped. Wellbore stability as drilling fluids that help maintain the integrity of the wellbore by counteracting formation stresses and preventing caving, sloughing, and swelling of reactive shales. This involves careful control of fluid density (hydrostatic pressure), chemical composition (to inhibit clay hydration), and filtration properties (to minimize pore pressure transmission) (Razzaq et al., 2021).

Formation pressure control is a critical mud system function by the action of the hydrostatic pressure exerted by the drilling fluid column must be sufficient to counterbalance the formation pore pressure to prevent fluid influx (kicks) and blowouts (Caenn et al., 2011). However, excessive hydrostatic pressure can lead to induced fracturing and lost circulation. Precise density control is therefore crucial for safe

operation. Bit cooling and lubrication as drilling generates significant heat at the bit. The circulating drilling fluid removes this heat, preventing thermal damage to the bit and downhole tools. Lubricating properties of the fluid reduce friction between the drill string and the wellbore wall, minimizing torque and drag. Drilling fluids can impact the accuracy of formation evaluation logs and the quality of formation samples (cuttings and cores). Minimizing filtrate invasion and ensuring compatibility with reservoir fluids are essential for obtaining reliable data, as the mud filtrate highly impacts the drilled formation petrophysics and geomechanics and might cause severe alterations to the formation characteristics (Gamal et al., 2021, 2020a).

Drilling Fluid Systems

The choice of drilling fluid system depends heavily on the geological environment, wellbore trajectory, temperature and pressure conditions, environmental regulations, and cost considerations (Caenn and Chillingar, 1996). Water-based fluids (WBFs) are the most common and cost-effective. Their base fluid is water, with various additives (clays, polymers, salts) to control rheology, filtration, and inhibit reactive formations. WBFs are generally environmentally friendly but can pose challenges in reactive shales and high-temperature, high-pressure (HTHP) environments. Oil-based fluids (OBFs) use diesel or mineral oil as the continuous phase, with water emulsified into the oil. OBFs offer excellent wellbore stability in reactive formations, superior lubrication, and HTHP performance. However, they are more expensive and have stricter environmental regulations. Synthetic-based fluids (SBFs) are similar in structure to OBFs but use synthetic oils as the base fluid. SBFs offer performance comparable to OBFs with improved environmental profiles, making them suitable for offshore operations.

Mud Invasion into Permeable Rock During Drilling: Mechanisms and Technical Consequences

The interaction between drilling fluid and the formation rock is a critical aspect of wellbore construction. During drilling operations, the pressure exerted by the drilling fluid column in the wellbore is typically maintained at a level exceeding the formation pore pressure. This pressure differential, known as overbalance pressure, is essential for preventing uncontrolled influx of formation fluids (kicks) and maintaining wellbore stability (Fernandes et al., 2013). However, this overbalance pressure also drives the movement of drilling fluid components, both liquid filtrate and solid particles, into permeable formations, as shown in Figure 1. This phenomenon, known as mud invasion or filtrate invasion, can have significant technical consequences that impact subsequent wellbore integrity, formation evaluation, and ultimately, hydrocarbon production.

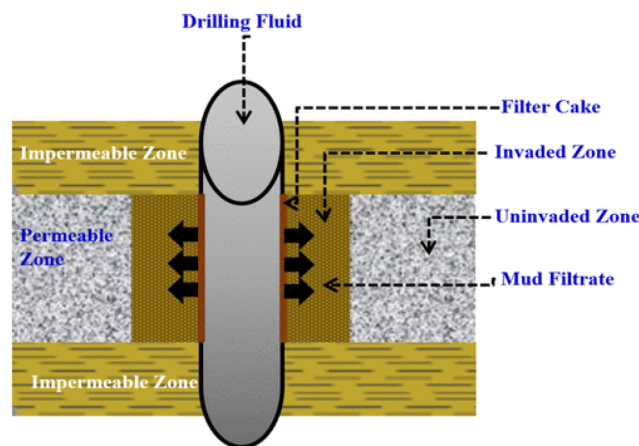


Figure 1—Drilling Mud Invasion for Permeable Zones (Gamal et al., 2020a)

Mud invasion is a complex process primarily governed by the pressure differential between the wellbore and the formation, and the properties of both the drilling fluid and the permeable rock. The liquid phase of the drilling fluid (filtrate) is the primary component that initially invades the permeable formation. As the drilling fluid contacts the permeable rock face, the overbalance pressure forces the filtrate into the pore spaces. This process continues until a filter cake is formed on the wellbore wall. As filtrate invades the formation, the solid particles suspended in the drilling fluid are deposited on the wellbore surface. These particles, if sized appropriately relative to the pore throats of the formation, will bridge across the pore openings and build a low-permeability layer known as the filter cake. The filter cake acts as a dynamic barrier that restricts further invasion of both filtrate and solid particles into the formation. While a well-designed filter cake effectively minimizes the invasion of most solid particles, some finer particles (e.g., clay fines, barite fines) may still penetrate into the near-wellbore region, particularly before a robust filter cake is fully established or if the filter cake is damaged. The extent of solid particle invasion depends on the particle size distribution of the drilling fluid solids, the pore throat size distribution of the formation, and the pressure differential.

Technical Workflow and Engineering Methodology

This section details the integrated technical workflow employed for successfully executing the combined off-bottom liner and lower completion operation in the high-permeability reservoir environment. The workflow (Figure 2) is structured into sequential phases, emphasizing data integration, advanced modeling, and dynamic operational management.

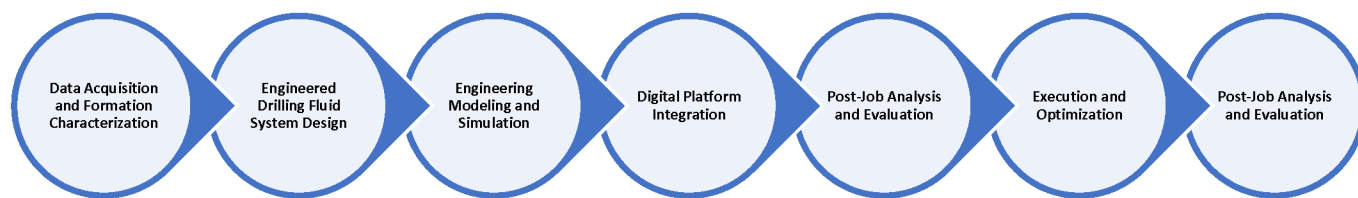


Figure 2—Technical Workflow and Engineering Methodology

Phase 1: Data Acquisition and Formation Characterization

This initial phase focuses on acquiring and analyzing all available data to build a comprehensive understanding of the drilling and well completion environment and specific reservoir characteristics relevant to the combined completion operation. Integration of diverse data types as wireline logs (porosity, permeability, lithology), core analysis data (pore size distribution, absolute and relative permeability, mineralogy), formation pressure test data, regional geological studies, and offset well performance data. Detailed permeability mapping to identify high-permeability zones and their lateral and vertical extent, with precise determination of reservoir pore size distribution (PSD) through core analysis and correlation with log data, as this provides the target for bridging material design. Identification of reactive minerals (e.g., clays) and their potential interaction with drilling fluids and the assessment of formation geomechanical properties relevant to wellbore stability.

Phase 2: Engineered Drilling Fluid System Design

Based on the formation characterization, this phase focuses on designing a drilling fluid system specifically tailored to mitigate the challenges posed by high permeability and the demands of a single-run combined completion. Defining key fluid performance criteria as effective bridging and sealing of high-permeability zones, minimal fluid loss (filtrate invasion), optimized rheology for hole cleaning, wellbore stability support, and non-damaging interaction with the reservoir.

Graphical particle size distribution matching through utilizing graphical techniques to compare the determined formation pore size distribution with the PSD of available bridging materials to select materials with appropriate size ranges to bridge the largest pore throats effectively (Dick et al., 2000; Mohamed, 2011). Employing engineering simulation software to model the filtration process under expected downhole pressure and temperature conditions as simulations are used to calculate the ideal blend and concentration of bridging materials required to achieve a target filter cake thickness and permeability, considering particle shape and packing efficiency. Also, selection of non-damaging bridging materials focusing exclusively on materials that are easily removed after the operation, with a strong emphasis on acid-soluble options to minimize residual formation damage.

Figure 3 displays the particle size distribution of the drilling fluid blend designed for bridging permeable formations. The x-axis shows particle size in microns on a logarithmic scale, covering a wide range from approximately 1 micron to 10,000 microns. The left y-axis indicates the volume percentage of particles within specific size bins, depicted by the blue "blend PSD" curve. This curve shows the concentration or distribution of particle sizes, highlighting the most prevalent sizes in the blend. The right y-axis represents the cumulative volume percentage, shown by the red "blend volume accum" curve. The vertical lines mark key percentiles of the cumulative distribution: d_{10} , d_{50} , and d_{90} that represent the particle sizes at which 10%, 50% (the median size), and 90% of the total particle volume is accumulated, respectively. Technically, this PSD data is crucial for designing bridging fluids, where the particle size range and distribution must be carefully matched to the formation's pore throat size distribution to effectively seal the formation face, minimize fluid loss, and prevent differential sticking.

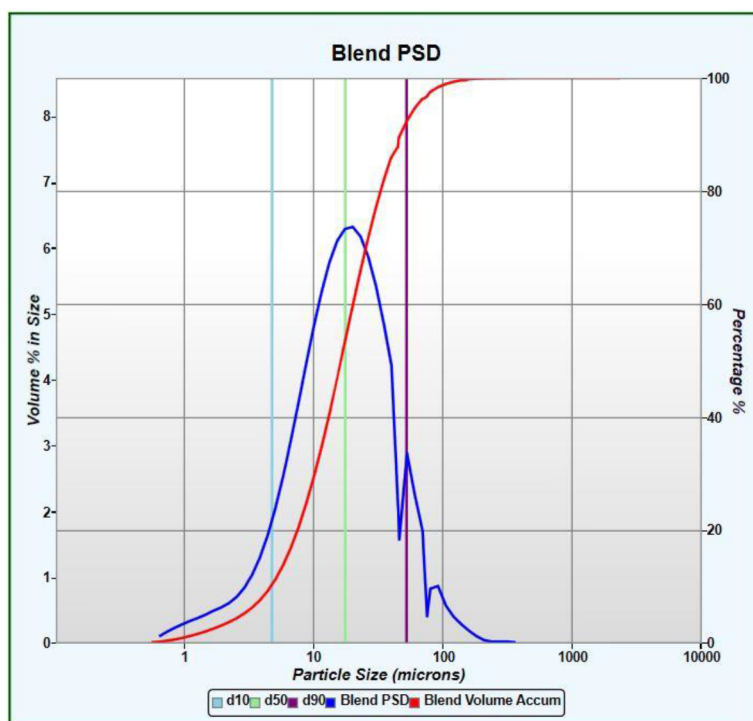


Figure 3—Particle Size Distribution (PSD) of Bridging Material Blend

Designing the fluid's rheological profile (Plastic Viscosity, Yield Point, Gel Strengths, Low-Shear-Rate Viscosity) to ensure effective cuttings transport in all wellbore sections, particularly deviated and horizontal intervals. Selecting high-performance fluid loss control additives (e.g., polymers, starches, cellulosic derivatives) to minimize filtrate volume and improve filter cake quality in conjunction with the optimized PSD.

Phase 3: Engineering Modeling and Simulation

Before execution, advanced engineering models are utilized to simulate the planned operation and predict potential outcomes, refining the plan for optimal performance and risk mitigation. Hydraulic modeling and simulation for the flow behavior of the drilling fluid in the annulus during tripping and circulation to predict equivalent circulating density and optimize pump rates and fluid properties for efficient hole cleaning and pressure management (Alharbi et al., 2025). Also, modeling the anticipated depth and volume of filtrate invasion into the permeable formation (up to 5000 mD) based on fluid properties, differential pressure, and exposure time, as this helps quantify potential formation damage risk (Gamal et al., 2020b).

Analyzing wellbore stresses and their interaction with fluid pressure and chemical activity to predict potential instability issues during the operation. Also, conducting comprehensive simulations that combine fluid hydraulics, wellbore mechanics, and completion string mechanics to evaluate the overall feasibility and potential risks of the single-run deployment under various scenarios. These models serve as benchmarks for evaluating field performance.

Figure 4 illustrates hookload simulations and actual measurements during drilling operations for analyzing drag and friction. Multiple lines represent simulated hookloads under various friction factors (FF), as the solid black line represents the Hookload as a reference, while scattered points show the actual measured hookload. By comparing the actual hookload data points to the simulated curves, engineers can infer the effective friction factor experienced in the wellbore. If the actual points fall within the range of simulated curves, it suggests a reasonable friction model. Deviations could indicate changes in wellbore conditions, fluid properties, or sticking issues. This type of analysis is crucial for understanding wellbore friction, optimizing tripping speeds, and identifying potential problems during drilling and completion operations.

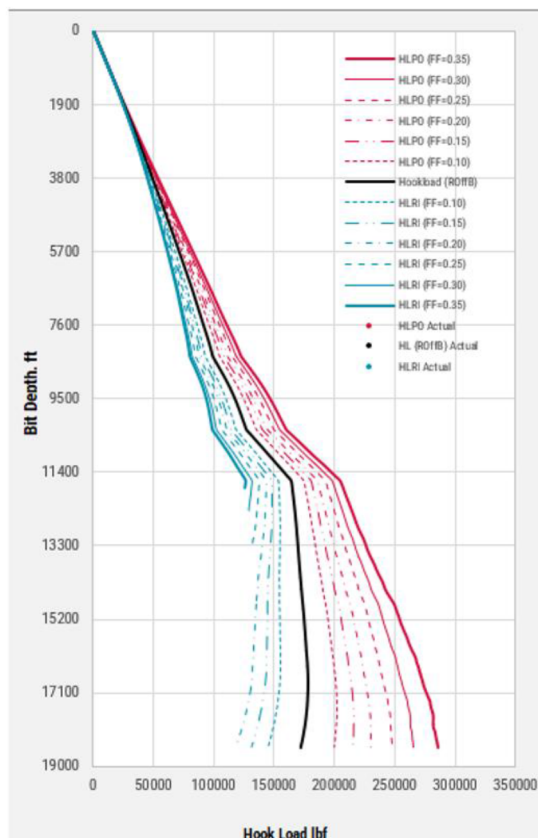


Figure 4—Simulated Hookloads Under Various Friction Factors

Phase 4: Digital Platform Integration

A digital platform serves as the central nervous system for data management and real-time monitoring throughout the workflow. Data acquisition and aggregation system to capture high-frequency real-time data from all relevant surface sensors (hookload, torque, RPM, pump rate, standpipe pressure, flow in/out) and downhole sensors, if available, and stream the data into a unified database within the digital platform. Creating intuitive dashboards within the digital platform to provide real-time visibility of key performance indicators (KPIs), fluid properties, and operational parameters for the engineering and operations teams. Figure 5 represents a synthetic sample from the real-time drilling digital platform, providing real-time operational data and performance analysis. The operating parameters section displays current drilling metrics like bit depth, ROP, Flow In, Rotary RPM, Hole Depth, WOB, SPP, and Torque. The hookload broomstick plot visualizes hookload versus depth, showing simulated curves for various friction factors and comparing them to actual hookload data points. This allows for real-time estimation of the effective friction factor. Similarly, the torque broomstick plot shows simulated torque profiles against depth with actual torque data points. The model vs actual section uses gauges to display the percentage deviation between modeled and actual torque and hookload, providing a quick visual of how well the model is predicting real-time conditions. This dashboard integrates real-time data with predictive models, enabling proactive monitoring of wellbore friction, early detection of potential issues like sticking or packing off, and optimization of drilling parameters for improved efficiency and safety.

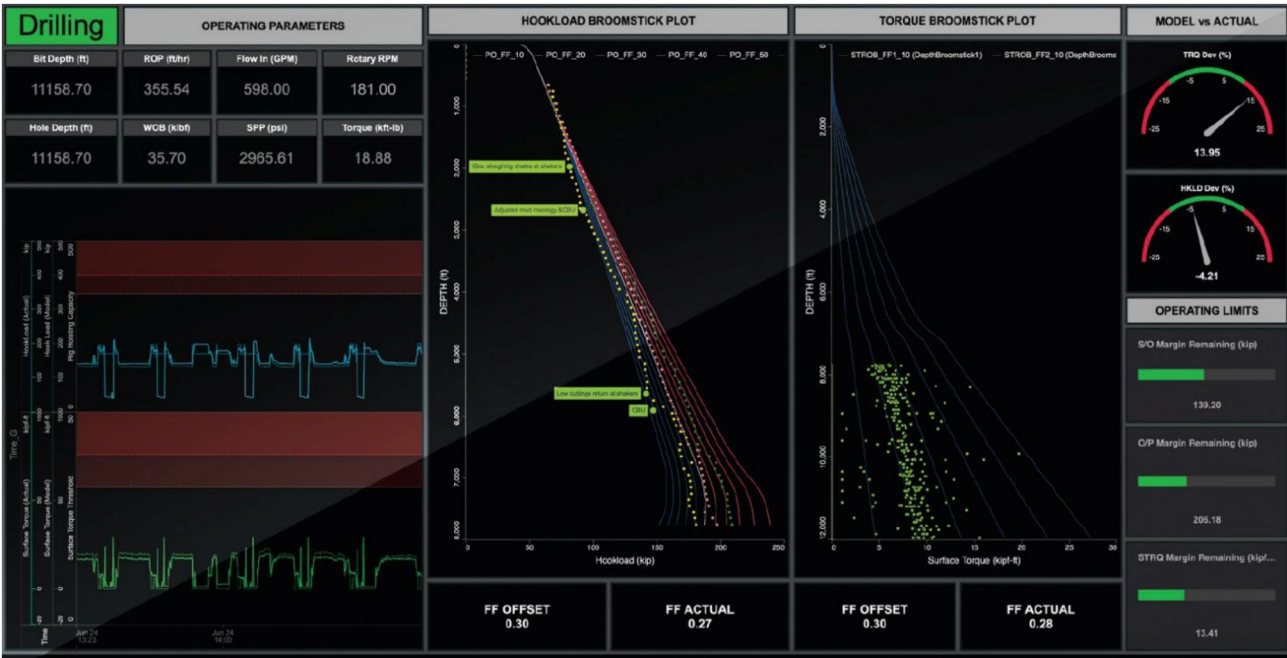


Figure 5—Drilling Digital Platform (Whitfield, 2022)

Phase 5: Operational Planning and Procedure Development

This phase translates the engineered design and simulation results into a detailed, actionable operational plan for the field crew. Following the engineering design and predictive modeling, a crucial phase focuses on translating these insights into a detailed operational plan for field execution, vital for complex combined completions. This phase begins by meticulously defining tripping speeds and circulation protocols. Recognizing risks like differential sticking and hole cleaning issues with a long, heavy string, guidelines are developed. Tripping speeds are optimized to manage surge and swab pressures, preventing wellbore instability. Circulation protocols define when and how to circulate during run-in, especially at critical depths, ensuring effective hole cleaning and preventing packing off, informed by hydraulic simulations.

Simultaneously, contingency planning is essential. Clear procedures and trigger points are established for potential issues like differential sticking, pressure surges, or losses, outlining immediate actions to regain control and mitigate problems. Furthermore, successful execution relies on clearly defined roles and responsibilities within the multidisciplinary team. Duties and authorities are explicitly outlined, with robust communication protocols ensuring seamless information flow and rapid decision-making. Decision-making authority for implementing contingencies is clearly defined. This integrated operational planning bridges engineering design and field execution, providing the field crew with a clear roadmap and necessary contingency measures for successful combined completion operations, contributing to precision and profitability.

Phase 6: Execution and Optimization

Following rigorous planning, the workflow transitions to execution and real-time optimization, the phase of physical well completion. This begins with pre-job fluid conditioning and quality control, where the drilling fluid is meticulously prepared at the surface to meet stringent design specifications, with comprehensive testing conducted before and during the operation. This phase of the operation involved running in hole with the integrated 4-1/2" off-bottom cemented liner and lower completion string as a single-trip system. The assembly included key lower completion components designed for selective isolation and inflow control screens. The entire string was deployed to the designed measured depth. Upon reaching the target depth, the wellbore was displaced to a solid-free water fluid system, ensuring proper fluid density and minimizing potential for formation damage prior to cementing. The combined liner and lower completion assembly was then successfully set and cemented in place, establishing zonal isolation and securing the string within the wellbore. Following the successful setting and cementing operation, the running tool was pulled out of the hole and subsequently laid down.

The core activity is running in the hole with real-time monitoring, deploying the combined string according to the planned protocols while continuously monitoring real-time data via the digital platform. This includes tracking torque, drag, pressures, and flow rates to assess progress and detect deviations as real-time insight from the digital platform, coupled with ongoing analysis from engineering models, facilitates dynamic parameter adjustment. Drilling parameters like pump rate or circulation strategy are adjusted proactively to maintain optimal performance and mitigate issues. The phase was accomplished by setting the liner and lower completion according to the engineered procedures. This dynamic, data-driven execution phase ensures the combined completion is installed efficiently and effectively, adapting to downhole conditions in real-time to achieve the desired outcome.

Phase 7: Post-Job Analysis and Evaluation

Upon completion of the combined well operation, the final phase, post-job analysis and evaluation, is critical for assessing performance and capturing valuable insights. This begins with a thorough performance comparison, where actual operational time and recorded non-productive time (NPT) are benchmarked against initial planned targets. A detailed evaluation of the drilling fluid's performance throughout the operation is also conducted, examining fluid properties, particle size distribution (PSD), and filtration control effectiveness. This involves analyzing post-completion well performance, primarily through production data, and reviewing relevant logs to identify any indications of formation damage or altered permeability. Finally, lessons learned and best practices are meticulously documented. This includes recording key observations from the planning and execution phases, identifying successes and challenges encountered, and capturing specific insights related to fluid performance, operational procedures, and the effectiveness of the digital platform and models. These documented lessons inform the refinement of future operational workflows and contribute to developing best practices for combined completion operations in similar challenging environments, driving continuous improvement.

Results and Observations

The implementation of the engineered workflow and integrated planning resulted in a highly successful combined well completion operation. The most significant outcome was the completion of the entire combined off-bottom liner and lower completion installation within 24 hours. This represented a 50% reduction in operational time compared to the conventional approach, which typically required 48 hours involving separate trips for the liner and the lower completion. All planned operational steps and activities were executed as designed, with no unplanned deviations from the procedure.

The drilling fluid successfully maintained an overbalance pressure of approximately ± 2000 psi throughout the operation while effectively managing permeability levels up to 5000 mD. Post-job evaluation indicated successful bridging and minimal fluid invasion into the high-permeability reservoir zones, suggesting the optimized PSD and filtration control were highly effective. The operation proceeded smoothly, with minimal indications of differential sticking or hole cleaning issues. The real-time monitoring through the digital platform provided clear visibility of operational parameters, enabling confident decision-making by the operational team. The seamless integration between the fluid design, operational plan, and execution highlighted the benefits of the collaborative engineering approach.

Conclusions

- The successful execution of this combined well completion operation directly validates the critical role of advanced drilling fluid engineering and integrated planning in overcoming complex technical challenges in high-permeability formations.
- The specifically engineered drilling fluid system, with its optimized PSD for bridging high-permeability zones and carefully controlled rheological and filtration properties, was fundamental to the success. The acid-soluble nature of the bridging materials ensured reservoir protection. The fluid's ability to effectively seal the formation under a significant overbalance pressure prevented differential sticking and minimized fluid invasion, directly addressing two major risks. The rheological design ensured effective hole cleaning throughout the long string deployment.
- The comprehensive planning phase, including detailed formation characterization, graphical and simulation-based PSD optimization, and predictive modeling, was essential. The simulations provided confidence in the fluid design and operational parameters, ensuring that the field execution was aligned with pre-defined performance expectations.
- The 50% reduction in operational time directly demonstrates the significant efficiency gains achievable through this single-run combined completion approach when supported by robust engineering. Eliminating a dedicated trip for the lower completion resulted in substantial rig time savings.
- The reduction in operational time directly translates into reduced project costs, as rig time is a major component of well construction expenses. Minimizing the risk of NPT associated with differential sticking or hole cleaning issues further contributed to cost efficiency.
- This case study provides compelling evidence that the single-run combined off-bottom liner and lower completions strategy is technically feasible and offers significant advantages in high-permeability environments, provided it is supported by advanced drilling fluid engineering and integrated planning.

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References

1. Alharbi, A., Petre, R., Gamal, H., 2025. Casing Exit Excellence: a Holistic Strategy of Integrated Engineering Models and Well Integrity Analytics for Extended Reach Wells.
2. Caenn, R., Chillingar, G.V., 1996. Drilling fluids: State of the art. *Journal of Petroleum Science and Engineering* **14**, 221–230. [https://doi.org/10.1016/0920-4105\(95\)00051-8](https://doi.org/10.1016/0920-4105(95)00051-8)
3. Caenn, R., Darley, H.C.H., Gray, G.R., 2011. Composition and Properties of Drilling and Completion Fluids, Composition and Properties of Drilling and Completion Fluids. <https://doi.org/10.1016/C2009-0-64504-9>
4. Dick, M.A., Heinz, T.J., Svoboda, C.F., Aston, M., 2000. Optimizing the Selection of Bridging Particles for Reservoir Drilling Fluids.
5. Fernandes, W., Jadhav, P., Zaki, M.M., Dutta, B., 2013. Integrated Drilling, Geosteering & Fluid Engineering Practices in Order to Maintain Horizontal Wellbore Stability & Maximize Reservoir Contact in a Channel Sand Reservoir.
6. Gamal, H., Elkatatny, S., Abdulraheem, A., 2020a. Effect of the Filtrate Fluid of Water-Based Mud on Sandstone Rock Strength and Elastic Moduli. *ACS Omega*. <https://doi.org/10.1021/acsomega.0c05067>
7. Gamal, H., Elkatatny, S., Adebayo, A., 2021. Influence of mud filtrate on the pore system of different sandstone rocks. *Journal of Petroleum Science and Engineering* **202**, 108595. <https://doi.org/10.1016/j.petrol.2021.108595>
8. Gamal, H., Elkatatny, S., Adebayo, A., Bageri, B., 2020b. Effect of exposure time on the compressive strength and formation damage of sandstone while drilling horizontal wells. *Journal of Petroleum Science and Engineering* **195**. <https://doi.org/10.1016/j.petrol.2020.107590>
9. Gamal, H., Elkatatny, S., Basfar, S., Al-Majed, A., 2019. Effect of pH on Rheological and Filtration Properties of Water-Based Drilling Fluid Based on Bentonite. *Sustainability* **11**, 6714. <https://doi.org/10.3390/su11236714>
10. Goud, M.C., Joseph, G., 2006. Drilling Fluid Additives and Engineering to Improve Formation Integrity.
11. Mohamed, M., 2011. Engineered Particle Size Distribution While Drilling Helped Minimize Wellbore Damage in Sandstone Reservoirs.
12. Mondal, S., Chatterjee, R., 2019. Quantitative risk assessment for Optimum Mud weight window design: A case study. *Journal of Petroleum Science and Engineering* **176**, 800–810. <https://doi.org/10.1016/j.petrol.2019.01.101>
13. Ong, G.S.H., Chan, L., Ho, A., Udin, R., Kunjirama, M., Nilpetploy, V., Thanasarnpisut, V., Duong, A., Azwar, C., 2024. Engineering Design Solutions of Drilling Fluid and Cement Slurry for a Challenging Deep HPHT Carbonate Appraisal Well in Malaysia.
14. Razzaq, W., Elkatatny, S., Gamal, H., Samsuri, A., 2021. The Utilization of Steelmaking Industrial Waste of Silicomanganese Fume as Filtration Loss Control in Drilling Fluid Application. <https://doi.org/10.1115/1.4051197>
15. Sheer, S., Karakaya, G., Gadhiya, K., Addagalla, A., Kamel, M., 2020. Engineered Fluid Design Helps Drilling the Longest Horizontal Section in Record Time-A Case Study from Turkey.
16. Whitfield, S., 2022. The art of data science: Industry hones in on reliability, ease of use. *Drilling Contractor*. URL <https://drillingcontractor.org/the-art-of-data-science-industry-hones-in-on-reliability-ease-of-use-63276> (accessed 6.11.25).